Lecture 28.06: Information Gain & Split Selection

📌 Topic Overview

How does a decision tree algorithm decide which feature gives the best split? The answer is **Information Gain (IG)**. In this lecture, we study the reduction-in-entropy formula and work through the exact numerical example taught on the blackboard.

🎯 What is Information Gain?

> **Information Gain measures the reduction in entropy (uncertainty) achieved after splitting a dataset based on a given feature.**

The intuition is simple:

$$extInformationGain = extEntropyofParent - extWeightedEntropyofChild$$

Mathematical Formula:

$$IG(S, A) = H(S) - \sum_{v \in ext{Values}(A)} ↑rac{|S_v|}{|S|} H(S_v)$$

Where:

- S = Parent dataset at the current node.
- $H(S)$ = Entropy of the parent node.
- A = Feature we are evaluating.
- v = A specific value or branch of feature A.
- S_v = Subset of samples where feature $A = v$.
- $rac|S_v||S|$ = Proportion of samples going down branch v (weight).

🧮 Blackboard Numerical Walkthrough

1. Parent Node Entropy

Consider the parent dataset with 6 samples: $[2y, 4n]$ (2 Yes, 4 No).

$$H(S) = - \frac{2}{6}\log_2\left(\frac{2}{6}\right) - \frac{4}{6}\log_2\left(\frac{4}{6}\right) \approx 0.918$$

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2. Candidate Feature 1: `Gender`

Splitting on `Gender` (Male vs Female):

- **Male branch (S_M)**: Contains 3 samples: $[2y, 1n]$.

$$H(S_M) = - \frac{2}{3}\log_2\left(\frac{2}{3}\right) - \frac{1}{3}\log_2\left(\frac{1}{3}\right) \approx 0.918$$

- **Female branch (S_F)**: Contains 3 samples: $[0y, 3n]$ (all No $\rightarrow$ Pure).

$$H(S_F) = 0.0$$

Now compute weighted child entropy:

$$\text{Weighted Child Entropy} = \frac{3}{6} \cdot H(S_M) + \frac{3}{6} \cdot H(S_F) = \frac{1}{2}(0.918) + \frac{1}{2}(0) = 0.459$$

Now compute Information Gain for Gender:

$$IG(extGender) = H(S) - 0.459 = 0.918 - 0.459 = 0.459$$

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3. Candidate Feature 2: `Employed`

The instructor notes on the board:

$$IG(extEmployed) = 0.800$$

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4. Split Selection Decision

We compare the Information Gain of all candidate features:

- $IG(extGender) = 0.459$
- $IG(extEmployed) = 0.800$

Since $0.800 > 0.459$, the tree selects `Employed` as the split feature at this node!

💻 Python Implementation

```
import numpy as np

def calculate_entropy(yes_count, no_count):
    """Calculates Shannon Entropy for binary classes."""
    total = yes_count + no_count
    if total == 0 or yes_count == 0 or no_count == 0:
        return 0.0
    p1 = yes_count / total
    p2 = no_count / total
    return -(p1 * np.log2(p1) + p2 * np.log2(p2))

def calculate_information_gain(parent_y, parent_n, left_y, left_n, right_y, right_
    """
    Calculates Information Gain achieved by splitting a parent node into two child
    Formula: IG = H(Parent) - [ (|Left|/|Parent|)*H(Left) + (|Right|/|Parent|)*H(R
    """
    total_parent = parent_y + parent_n
    total_left = left_y + left_n
    total_right = right_y + right_n

    h_parent = calculate_entropy(parent_y, parent_n)
    h_left = calculate_entropy(left_y, left_n)
    h_right = calculate_entropy(right_y, right_n)

    weighted_child_entropy = (total_left / total_parent) * h_left + (total_right /
    info_gain = h_parent - weighted_child_entropy
    return info_gain

# --- Execute Calculation on Classroom Video Example ---
# Parent Node: [2 Yes, 4 No]
parent_entropy = calculate_entropy(2, 4)
```

```
# Candidate 1: Split on Gender -> Male [0y, 3n], Female [2y, 1n]
ig_gender = calculate_information_gain(2, 4, 0, 3, 2, 1)

# Candidate 2: Split on Employed -> Employed [2y, 0n], Unemployed [0y, 4n]
ig_employed = calculate_information_gain(2, 4, 2, 0, 0, 4)

print(f"Parent Node Entropy:         {parent_entropy:.3f}")
print(f"Information Gain (Gender):   {ig_gender:.3f}")
print(f"Information Gain (Employed): {ig_employed:.3f}")
print("Best Split Selected:", "Employed" if ig_employed > ig_gender else "Gender")
```

🔑 Key Takeaways

1. Information Gain represents the reduction in entropy caused by a split.
2. The feature with the **highest Information Gain** is chosen as the split node.
3. When using Gini Impurity, the equivalent metric is **Gini Gain**.